

Even Keel Learning

A note for Laura

From Stuart Rainey · May 2026

Laura — you're a teacher and a parent, so I've written this for both halves of the same person. You've probably already heard twenty pitches for "the AI homework tool that won't cheat." So have I. This isn't another one of those. The whole point of what I've built is that it works **without** an AI in the part that talks to your child, and that's the bit nobody else in the market is doing.

Here's what it actually is, in one paragraph.

****Even Keel is a homework helper for ages 11–18 that never gives the answer.**** It runs in your child's browser. It teaches the way a good tutor does — three small hints, then a worked example with different numbers, then *"now you try"*. There is no chat-bot under the bonnet, so there is nothing to *make up an answer* and nothing to *hallucinate*. Safeguarding is built into the engine, not bolted on afterwards. And every claim I make about it can be checked — there's a public "honesty document" that lists what's real and what isn't.

What you'd see as a teacher

You already know the way every existing AI homework tool fails: the child types the question in, copies the reply, ticks the box, and learns nothing. You can't trust the work, can't grade it fairly, can't even tell if they understood any of it.

Even Keel was designed to remove that path completely.

- **It cannot give the answer.** Not "won't" — *can't*. There's no AI inside it to generate one. Hints come from a small hand-written library, like a teacher's prompt cards.
- ****It recognises the *kind* of mistake**** the child is making.

If a Year 9 student flips a minus sign or makes an off-by-one slip, the engine notices the *category* of error and asks a Socratic question pointed at that mistake — without ever showing the corrected number on screen.

- **There's a "now I'll do one with you" move.** When three hints aren't enough, the platform serves a fully worked example — same shape, different numbers — walks through it end to end, then hands the original problem back. Every decent tutor does this. Other AI tools just give the answer.

- **You'd see a live "Integrity Ledger".** As your class works at home, you'd see — per child — how many hints they needed, which comprehension questions they cleared, whether they pasted anything in, and whether the engine flagged anything. **You'd never see what they typed.** Just the shape of how they worked.

- **Comprehension Gate.** Before a topic counts as "done", the child has to answer a few questions about *why* the method works, not just whether they got the number right. Wrong answers come with explanations and a chance to try again — never the answer.

- **Signed receipts.** When a child finishes a problem, they

can issue a small "receipt" — a link they share with you — that records what was solved, how many tries, which hints, how long. You open the link, click *Verify*, and the page confirms the receipt hasn't been edited since they made it. No log-in. No account. It just works in your browser.

A straight piece of disclosure on subject coverage, because you'll spot this within thirty seconds of clicking around. The student page has a **subject picker showing about 60 subjects** — Maths, English Lit, Physics, French, Religion / RE, Welding, all the way through to Photography. That picker is the *shape* of where the platform is heading; it is not yet the *content*. ***Today, real validated content lives in one skill family — linear equations.*** Click any other tile and the title bar changes ("*ENGLISH-LIT · today's problem*") but the problem you get is the same Maths equation. I'd rather you walk in knowing that than discover it during a demo.

The reason it isn't *"just generate questions for English Lit with an LLM"* is that the moment I do that, every promise on the previous page (no answer-generation code path, no hallucination risk, provable-by-**grep** safety) collapses. So each new subject takes hand-written content: a checker (or an explicit "qualitative — no auto-check" mode), a small library of worked parallel problems, and three to five Comprehension-Gate questions per topic. A specialist who knows the subject can write that in roughly a week per skill family. It is content work, not engineering work — which is actually a good place to be.

For **essay-style work** specifically I am not willing to ship an automatic grader — if a tool can't mark an essay reliably, it shouldn't pretend to. What you *can* use the platform for in an English context today is the **Socratic engine** for analysis questions you author yourself, the **safeguarding pipeline**, the **comprehension and reasoning checks**, and the **paste-and-cheat defences** — all of which apply to any homework, not just numeric. Subject-specific content is the next phase, and a pilot is exactly the right scale at which to write the first round of it together.

What you'd see as a parent

Same product, different reading.

- **Nothing your child types leaves the device.** The platform

doesn't have a server it sends conversations to. The chat itself lives in the browser's memory only — close the tab and it's gone. There is no library of "your child's chats" I could ever publish, leak, or sell, because no such library is ever created.

- **A weekly snapshot, never a CCTV feed.** What's shared with

you and the teacher is *categorical*: the *shape* of how your child worked, never the words they used. *"Sam asked for two hints on quadratics and cleared the comprehension check"* — that level. Not a transcript.

- **No paid model, no in-app upsell, no harvested data.**

Because there's no back-end, there's nothing to harvest into. This isn't a privacy promise; it's the current shape of the thing.

- **A guardian-acknowledgement step for under-13s.** Before a

child under 13 can use it, an adult has to tick a box. (I'm honest that this is a tick-box today and the next phase upgrades it to verified parental consent — email confirmation or school-mediated.)

- **Built for the broadest set of children.** The platform is

designed to the accessibility standard schools work to (WCAG 2.2 AA), in nine languages including Arabic and Irish. One automated check (colour-contrast) is currently scoped out of the test suite while the design system is finished — logged honestly in the changelog, not hidden. There's a speech-to-text mode for

severe dysgraphia, a dyslexia font option, and a special path for children using switches, eye-gaze, or dictation so they aren't penalised for steady keystrokes.

- **The honesty document is for parents, not lawyers.** It's

written so an attentive parent can read it and understand what's real and what isn't.

A Tuesday night, kitchen table

Your child opens the app. They pick a tone of voice — *mentor* (calm and patient), *peer* (sounds like the kid sat next to them), or *foreman* (no-nonsense, like a site lead). They start a Maths problem.

They get stuck. They tap **Hint**. The app says "start by isolating the variable on one side." They get stuck again — Tier 2 — "what does multiplying both sides by minus-one do to the inequality direction?" They have a go, get a sign wrong, type their answer. The app spots the sign-flip and says "check the sign on the right — what's the rule when you multiply by a negative?" — without ever showing the right number.

They're still stuck. Tier 3. Still stuck. The app shows them a **different problem of the same shape**, walks through it from start to finish, then hands the original back. They solve it. They tap *Issue receipt*. A short link is in your messages by the time tea is on the table.

If at any point your child had typed "I don't want to wake up tomorrow", the platform would have caught it before replying anything else. It would have shown the Childline number, refused to give a hint, and quietly notified the school's safeguarding inbox — **without ever recording the words your child wrote**. Only the *category* of the alarm. The platform supports the school's KCSIE and Prevent duties without the school taking custody of text it doesn't want — the school's own policies and the DSL still have to do their part.

Who else this helps

The classroom is the front line, but the same design holds up for everyone else with a stake in what your child is doing.

- **The school, the Head, the governors.** I produce a single

signed document — the *transparency bundle* — that wraps the safeguarding controls, the policy mappings, and a record of exactly what's in the codebase into one file. The school's data protection officer can hand it to a regulator. The regulator can check it themselves, in their own browser, with no log-in.

- **The Multi-Academy Trust / Board of Education.** They can

compare this transparency bundle to whatever any other AI vendor is offering and make a real apples-to-apples procurement decision. Most are still being asked "is this safe?" and getting back forty-page brochures.

- **The Designated Safeguarding Lead.** The crisis pipeline is

category-only by construction — the words a child typed cannot reach the safeguarding inbox even by accident, because the channel that carries the alert literally has no field for them. The DSL can also fire a synthetic test alert any time, to confirm the pipeline is alive without waiting for a real incident.

- **Regulators (Ofsted, Ofqual, ICO).** Everything is checkable

offline, in a browser, with no vendor account.

- **Exam boards, in time.** The signed receipts are designed

with coursework provenance in mind. Today, one teacher accepting one receipt for one piece of work is the smallest case. Tomorrow, an exam board could accept a receipt to attest "this coursework was produced

through this Socratic process, with this many hints, in this much time"* — not "this child wrote it", but a much richer signal than a single graded paper.

- **The PTA / parent community.** A document parents can read

end-to-end and a platform with no data harvesting to argue about.

Three questions you raised

"Does it track how long was spent on homework?"

Per problem, yes. Per assignment or team assignment, not yet. Every problem session records start time, finish time, total think-time between keystrokes, deletions, and the steps the child took. What the platform doesn't yet have is the concept of an *assignment* (a homework set of several problems) or a **team assignment** (children working together) — those need a back-end which is the next phase. The smallest honest fix in the meantime is to surface time-on-task to teachers and add it to the receipt — that's about half a day of work and I can ship it for the pilot.

"Can it tell if an answer was pasted from another AI?"

****It detects that a paste happened. It does not — and does not claim to — prove the source was an AI.**** The chat box blocks pastes by default and counts every attempt. The "trust meter" combines paste pressure with steady-cadence typing ("typing along while reading another tab") and tab-switching ("alt-tab, paste, alt-tab back") into a green/amber/red light. That's *behavioural* evidence, not forensic identification. I deliberately don't ship an "AI detector" — every credible study says the current generation of detectors are wrong often enough to wrongly accuse real students, and a wrongly-accused child is the kind of incident that ends a pilot.

"What about banning copy-paste — for every situation?"

The honest position: ****block it where it doesn't belong, count it everywhere, accuse never.**** Copy-paste is an integrity tool, not a discipline tool. By scenario:

- **Maths or short-answer homework on the chat rail.** Block, hard. This is the exact fraud the rail is designed against. Already shipping.

- **Essay or long-form writing.** Don't block — children draft

in Word and paste in, that's normal. Capture the size of the paste and the pattern around it as a *signal* the teacher can see, not a verdict the platform makes.

- **Programming homework.** Don't block. Pasting between files is the normal workflow. Different problem, different module.

- **Citations and quotes.** Allow but route through a "paste as quote" button so it's clear the child knows it's not their own writing.

- **Children using dictation, switches, or eye-gaze input.**

Already handled correctly: steady-cadence typing isn't treated as suspicious for them (because it's *normal* for them), but pasting is still counted because pasting is a deliberate action whatever the input device. The message should be gentler though — **"we noticed a paste; that's fine, just not part of your work"** rather than a red flag.

- **Younger children (under 13).** Block by default and frame

the trust meter as a *green-light* signal, never as surveillance.

- **Formulas or constants.** Allow short pastes; flag large

blocks. Length matters more than count.

The principle: the platform should always *count* pastes; it should only *block* them where pasting is structurally incompatible with the lesson. Anywhere else, surface the signal in context and let the teacher make the call.

What it doesn't do today

I keep this list short on purpose. It's also the reason the rest of the document is trustable.

- **No AI in the bit that talks to your child.** That's the point, not a gap.

- **No back-end yet.** Pilot integration is the next phase. A single-school pilot can run against a webhook the school controls.

- ****No real text/email/SMS notifications to the safeguarding team yet.**** The school's webhook is real; email/SMS/push delivery to a named provider needs a paid account and a school billing relationship.

- **The under-13 age check is currently a tick-box.** Next phase: verified parental consent.

- **Essay grading is deliberately not built.** Until it can mark reliably, it shouldn't.

Every one of these is in the public honesty document.

What a pilot with you could look like

The platform is designed around the smallest possible useful shape:

1. One subject, one year group, one class. (Maths Year 9 is the strongest fit today; for English, the comprehension and safeguarding pieces apply equally.)
2. Two weeks of homework only. Children use it at home; you check the Integrity Ledger in the morning.
3. One signed receipt per completed problem. At the end of the pilot, you accept (or decline) one piece of work purely on the basis of a receipt — the smallest **"the system was trusted to attest something"** event.
4. One safeguarding rehearsal. With school consent, the DSL fires a synthetic alert and confirms it lands in the school's chosen inbox.
5. One transparency-bundle hand-off to the school's data protection officer at the end.

Next step

If this looks worth your time, the simplest next step is a 30-minute screen-share where I walk you through:

1. The student rail — one Maths problem, three hints, the parallel, the receipt.
2. What the teacher view looks like as the same problem is being solved.
3. The safeguarding side — one synthetic alert, signed and verified end to end.
4. The transparency bundle — the file your school's DPO would actually receive.

Everything in those four steps is shipping today and reproducible from a clean install on your laptop, not a marketing demo.

The honesty document this note is built on is called HONESTY.md and lives in the codebase. Nothing in this note claims anything that isn't sitting in that file.